

Zhixuan Liu

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Research Interests: LLM training and inference, with a focus on safe distillation, inference-time alignment, efficient decoding, and search-based generation

Education

Shanghai Jiao Tong University	Sep. 2022–Jun. 2027 (Expected)
Ph.D. Candidate in Computer Science; School of Electronic Information and Electrical Engineering	Shanghai, China
Shanghai Jiao Tong University	Sep. 2018–Jun. 2022
B.Eng. in Computer Science; ACM Class, Zhiyuan College	Shanghai, China

Research Experience

Shanghai Artificial Intelligence Laboratory	Research Intern, 2022–Present
<i>Safe distillation, inference-time alignment, and efficient LLM decoding</i>	

Subliminal Learning as Trait-Direction Drift: A Mechanism and Targeted Control under SFT Distillation *First Author; Ongoing Work*

- Independently led the problem formulation, method, implementation, experiments, and writing; introduced trait-direction drift to explain how latent traits transfer through semantically unrelated distillation data.
- Developed probe-space corridor regularization, reducing malicious-response transfer from **29.55%** to **6.47%** while preserving task performance.

Decoupled Contrastive Decoding via Expert-Aligned Drafting (DCD) *First Author; Ongoing Work*

- Independently led the method, SGLang implementation, experiments, and writing; replaced serial amateur proposals with an expert-aligned drafter while preserving the original contrastive-decoding distribution.
- Achieved **1.65–1.95×** average speedups across three 8B settings on HumanEval, GSM8K, MMLU, and CNN/DM, scaling to approximately **2×** at 70B.

Inference-Time Language Model Alignment via Integrated Value Guidance (IVG) *Co-first Author (First-listed); EMNLP 2024 Findings*

- Led the problem formulation, method, code, experiments, and writing; combined token-level implicit values with chunk-level search to align frozen target LLMs at inference time.
- Raised AlpacaEval 2.0 LC win rates for Mistral-7B / Mixtral-8×7B from **19.51% / 25.58%** to **26.51% / 33.75%**; extended IVG to SafeWork-R1, improving safety score from **77.1** to **93.8**. [\[Code\]](#)

Weak-to-Strong Search: Align Large Language Models via Searching over Small Language Models *Second Author / Core Contributor; NeurIPS 2024*

- Contributed deeply to the idea, method, implementation, experiments, and writing; used small-model preference differences as dense rewards and values for chunk-level search over frozen LLMs, including cross-tokenizer and black-box settings.
- Using a 7B weak model with about **9.5%** LC win rate, improved Llama-3-70B-Instruct from **34.42%** to **39.09%** and black-box GPT-3.5-Turbo-Instruct from **16.00%** to **20.07%**. [\[Code\]](#)

ByteDance AI Lab **Research Intern, Sep. 2020–Jun. 2021**

Neural-symbolic learning, program synthesis, and numerical reasoning

A Neural-Symbolic Approach to Natural Language Understanding *First Author; EMNLP 2022 Findings*

- Led the problem formulation, method, implementation, data, experiments, and writing; proposed a Mixture-of-Experts Neural-Symbolic Processor combining neural prediction with program generation and symbolic execution.
- Built **32,011** DROP program annotations and **722** AWPNNLI E/C-Program examples; reached **80.81 EM / 84.01 F1** on DROP and **92.24%** accuracy on AWPNNLI. [\[Code\]](#)

Other Selected Publications

<i>Native Reasoning Models: Training Language Models to Reason on Unverifiable Data</i>	ICLR 2026 Poster; Second Author
<i>Adaptive Prompt Optimization for Open-Ended Tasks: Uncertainty Preference as a Secondary Signal</i>	ACL 2026 Findings; Second Author
<i>Emergent Response Planning in LLMs</i>	ICML 2025 Poster; Third Author

Skills & Honors

Programming and Frameworks: Python, PyTorch, Hugging Face Transformers, SGLang **Language:** English (CET-6)
Core: Model distillation and SFT, DPO, reward/value modeling, speculative/contrastive decoding, search-based generation
Honor: Silver Medal, National Olympiad in Informatics (NOI), 2017